That is, a square matrix $A=[a_{ij}]_{n\times n}$ is said to be a scalar matrix if $a_{ij}=\begin{cases} c & \text{if } i=j \\ 0 & \text{if } i\neq j \end{cases}$ where c is a fixed number. For instance,

$$A=\begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix}, B=\begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}, C=\begin{bmatrix} \sqrt{3} \end{bmatrix} \text{ and } D=\begin{bmatrix} c & 0 & \cdots & 0 \\ 0 & c & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c \end{bmatrix}\begin{matrix} R_1 \\ R_2 \\ \\ R_n \end{matrix}$$

are scalar matrices of order 3, 2, 1, and n respectively.

Observe that any square zero matrix can be considered as a scalar matrix with scalar 0.

Definition 7.9

A square matrix in which all the diagonal entries are 1 and the rest are all zero is called a **unit matrix**. Thus, a square matrix $A=[a_{ij}]_{n\times n}$ is said to be a unit matrix if $a_{ij}=\begin{cases} 1 \text{ if } i=j \\ 0 \text{ if } i\neq j \end{cases}$.

We represent the unit matrix of order n as I_n. For instance,

$$I_1=[1], I_2=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ and } I_n=\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}\begin{matrix} R_1 \\ R_2 \\ \\ R_n \end{matrix}$$

are unit matrices of order 1, 2, 3 and n respectively.

Note 7.1

Unit matrix is an example of a scalar matrix.

There are two kinds of triangular matrices namely upper triangular and lower triangular matrices.

Definition 7.10

A square matrix is said to be an **upper triangular matrix** if all the elements below the main diagonal are zero.

Thus, the square matrix $A=[a_{ij}]_{n\times n}$ is said to be an upper triangular matrix if $a_{ij}=0$ for all $i>j$. For instance,

$$\begin{bmatrix} 4 & 3 & 0 \\ 0 & 7 & 8 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 0 & 1 \end{bmatrix}, \text{ and } \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & a_{nn} \end{bmatrix} \text{ are all upper triangular matrices.}$$

Definition 7.11

A square matrix is said to be a **lower triangular matrix** if all the elements above the main diagonal are zero.

More precisely, a square matrix $A=[a_{ij}]_{n\times n}$ is said to be a lower triangular matrix if $a_{ij}=0$ for all $i<j$. For instance,

07_11th-Maths-Vol2-Unit7.indd 6 23-03-2019 09:58:44